SafeRoute BLR

Comprehensive Project Architecture, Algorithms & Data Flow Specification

v2.0 Desktop Suite

Production Ready

Target Region: Bengaluru Metropolitan Area

Backend: Python 3.12 (REST HTTP Service)

Frontend: Google Maps JS API + Route Safety Studio

Intelligence: NCRB + BCP + OpenStreetMap Proxies

1. Executive Summary & Core Value Proposition

Conventional urban navigation systems (such as standard GPS devices and commercial routing engines) optimize routes strictly for **minimal distance or shortest travel time**. In major metropolitan areas like Bengaluru, this often funnels pedestrians, two-wheelers, and late-night commuters through dimly lit alleys, isolated ring road bypasses, or areas with sparse emergency service coverage.

SafeRoute BLR bridges this safety gap by evaluating each candidate corridor across an empirical **5-Factor Safety Index**. The system synthesizes official National Crime Records Bureau (NCRB) crime statistics, Bengaluru City Police jurisdiction coordinates, spatial street lighting coverage, commercial crowd presence, transit connectivity, and dynamic time-of-day risk fluctuations to grade road options into clear **Green (Full Safety)**, **Yellow (Medium Safe)**, and **Red (Not Safe)** tiers.

2. Tool Stack & Component Breakdown

Component	Technology Stack	Functional Role & Core Architecture
`desktop_app.py`	Python 3.12, `http.server`, `threading`, `hashlib`	Desktop application backend and window manager. Runs a multi-threaded HTTP server with automatic port discovery (8050+), provides local REST APIs (`/api/bootstrap`, `/api/auth/*`, `/api/analyze-routes`, `/api/csv`), enforces PBKDF2-SHA256 password security, and spawns the UI in native desktop application mode.
`desktop_ui.html`	HTML5, Modern CSS Grid/Flexbox, ES6+	Desktop presentation layer featuring a professional Google Email ID & password sign-in gate, tabbed workspace (Live Google Map, Route Safety Studio, NCRB Crime Intelligence), interactive 5-factor gauge inspector, and a bidirectional `postMessage` synchronizer with the embedded map engine.

Component	Technology Stack	Functional Role & Core Architecture
`bangalore_map.html`	Google Maps JS SDK, OSRM API, FontAwesome	Full-featured geospatial client supporting Roadmap, Satellite, and Terrain views. Manages multi-route corridor discovery, dynamic polyline rendering (highlighting and dimming), double-click single route selection, live police station markers, HTML5 geolocation permissions, and a floating live GPS HUD.
`bengaluru_crime_extractor.py`	Python 3, `pdfplumber`, `pypdfium2`, `pandas`	Crime intelligence data pipeline that ingests official NCRB "Crime in India" multi-page PDF documents and extracts 308 verified tabular datasets for Bengaluru into standardized CSV files with summary JSON metadata.
`launch_desktop.bat` & `SafeRoute BLR.lnk`	Windows Batch & Shell Shortcut	One-click application launcher that detects virtual or system Python environments, binds high-resolution icon assets, and bootstraps the desktop environment seamlessly without manual terminal commands.
`assets/`	SVG, PNG, Multi-size ICO	Branding package containing vector shield/skyline graphics, 512px raster assets, and 16px–256px icon binaries embedded into desktop shortcuts and taskbars.

3. Mathematical Formulations & Algorithmic Models

3.1. The 5-Factor Weighted Safety Score

The safety score $S \in [0, 100]$ along any road corridor is formulated as a linear combination of five normalized safety factor functions $f_k \in [0, 1]$ weighted by time-of-day coefficients w_k:

```
S = 100 × [ w_crime · (1 − c) + w_police · (1 − p_dist) + w_lighting · l
+ w_crowd · f_crowd + w_isolation · (1 − i) ]
```

1. Low Crime Safety (1 - c)

Derived from locality incident baselines and NCRB metropolitan records. High incident density increases crime index c, lowering safety points.

2. Police Station Proximity (1 - p_dist)

Measures corridor closeness to the nearest Bengaluru police station: $p_{\text{dist}} = \min(1.0, d_{\text{min_police}} / 5.0\text{ km})$. Routes within 500m score >90%.

3. Street Lighting Coverage (l)

Quantifies street lamp density and illumination across sampled road coordinates. Critical for preventing unlit blind spots.

4. Crowd & Open Shops (f_crowd)

Gauges human footfall, active commercial retail establishments, and transit nodes (Metro/BMTC stops) providing natural surveillance.

5. Area Connectivity / Low Isolation (1 - i)

Penalizes remote, desolate, or walled stretches with low street egress. Arterial ring roads and populated avenues maintain near-zero isolation ($i \to 0$).

3.2. Dynamic Time-of-Day (ToD) Reweighting

Factor Name	Daytime Weight (2:00 PM)	Nighttime Weight (11:00 PM)	Behavioral Rationale
Low Crime Safety	**35%** (0.35)	**30%** (0.30)	Baseline crime history remains influential across all hours.
Street Lighting	**20%** (0.20)	**28%** (0.28)	Illumination becomes a dominant safety determinant after sunset (+8% boost).
Crowd & Open Shops	**15%** (0.15)	**20%** (0.20)	Open commercial establishments provide vital "eyes on the street" at night (+5% boost).
Police Station Proximity	**20%** (0.20)	**12%** (0.12)	Proximity remains steady, balanced with immediate street lighting needs.

Factor Name	Daytime Weight (2:00 PM)	Nighttime Weight (11:00 PM)	Behavioral Rationale
Low Isolation (Connected)	**10%** (0.10)	**10%** (0.10)	Ensures routes maintain multiple alternative exit points and arterial road access.

3.3. Safety Categorization Bands

Band Range	Visual Badge	Display Label	Traveler Guidance
75.0% – 100.0%	**Green (#16a34a)**	**Full Safety**	Recommended corridor. High lighting, heavy footfall, minimal isolation, and immediate police accessibility.
50.0% – 74.9%	**Yellow (#d97706)**	**Medium Safe**	Acceptable for daytime travel. Exercise caution late at night or during off-peak hours.
0.0% – 49.9%	**Red (#dc2626)**	**Not Safe**	High-risk corridor due to sparse lighting, high historical incident density, or severe isolation.

4. Spatial Geometry & Navigation Algorithms

4.1. Great-Circle Distance (Haversine Formula)

Used for point-to-point corridor sampling, nearest police station matching, and live off-route drift tracking:

$$a = \sin^2(\Delta\varphi/2) + \cos(\varphi_1) \cdot \cos(\varphi_2) \cdot \sin^2(\Delta\lambda/2) \quad c = 2 \cdot \operatorname{atan2}(\sqrt{a}, \sqrt{(1-a)}) \quad d = R \cdot c \quad \text{(where } R = 6{,}371 \text{ km)}$$

4.2. Multi-Route Exploration & Selection State Machine

To give travelers complete clarity without visual overload, SafeRoute BLR implements a 3-state route selection lifecycle:

1. **Initial State: All Routes Shown (`activeDirRouteIdx = -1`)**
   When the routing engine computes alternatives, all candidate corridors are rendered at full standard visibility (5.5px stroke, 8px halo, 0.94 opacity, 1.0 badge opacity). No route is pre-dimmed, allowing the user to view all paths side-by-side.
2. **Single-Click: Route Highlighting & Background Dimming (`activeDirRouteIdx = idx`)**
   Clicking any route line, marker badge, or sidebar card elevates the selected corridor (7.5px line, 12px white halo, 1.0 opacity, z-index 60, badge z-index 100) and dims all alternative corridors (3.5px line, 0.28 opacity, faint halo, badge opacity 0.40). The Route Studio highlights the matching card and factor gauge.
3. **Double-Click: Single Route Locked & Alternatives Cancelled**
   Double-clicking locks that specific corridor to reach the destination and cancels all other alternatives (`dirRoutes = [chosenRoute]`). Alternative polyline overlays and badges are removed from the map. A "Show All / Restore Alternatives" button enables one-click restoration.

4.3. Live GPS Destination Tracker & Off-Route Detection

Continuous GPS Tracking

Utilizes `navigator.geolocation.watchPosition` with high accuracy mode, plotting a live blue marker, dynamic accuracy radius circle, and polyline breadcrumb trail.

Off-Route Detection (> 0.35 km)

Calculates minimum perpendicular distance from live coordinate (lat_u, lng_u) to route segments. If drift exceeds 350 meters, the HUD triggers an orange off-route warning and offers one-click recalculation.

Arrival Detection (≤ 0.06 km)

When remaining distance to destination falls within 60 meters, the HUD status turns green with "Arrived at Destination safely".

Floating Navigation HUD

Displays real-time remaining distance (km), dynamic ETA based on speed, progress percentage bar, current corridor safety score %, and closest police station name with distance.

5. Security, Cryptography & Data Synchronization

5.1. Authentication Architecture

SafeRoute BLR features a dedicated desktop login page enforcing Google Email ID identification with secure local password hashing:

- **Domain Enforcement:** Rejects non-Google webmail domains (Yahoo, Outlook, Proton, iCloud, Zoho). Permits `@gmail.com`, `@googlemail.com`, and corporate Google Workspace domains.
- **Password Storage:** Passwords are never stored in plaintext. They are hashed using **PBKDF2-HMAC-SHA256** with **120,000 iterations** and a 16-byte random salt generated by `secrets.token_bytes(16)`.
- **Timing-Attack Defense:** Password hashes are validated using constant-time string comparison (`hmac.compare_digest`).
- **Session Management:** Active session cached in `data/user_session.json` when "Keep me signed in" is checked; multiple accounts saved in `data/auth_accounts.json` for instant switching.

5.2. Two-Way `postMessage` Communication Bridge

Because the Google Maps client runs inside an embedded iframe within the desktop UI, a robust message passing protocol keeps both interfaces synchronized:

Message Type	Source → Target	Payload & Synchronized State
`blr-sync-directions`	Studio → Live Map	Origin, Destination, Travel Mode, Time-of-Day. Triggers map route calculation.
`blr-directions-updated`	Live Map → Studio	Calculated route summaries, distance, duration, and 5-factor safety scores. Populates Studio cards.
`blr-select-route`	Studio → Live Map	Route index `idx`. Highlights route and dims other routes on map.
`blr-route-selected`	Live Map → Studio	Route index `idx`. Updates active card and factor gauge in Studio.
`blr-lock-single-route`	Bidirectional	Locks chosen corridor, removes all alternative routes, displays confirmation banner.
`blr-routes-restored`	Bidirectional	Restores all discovered alternative routes back to equal visibility.
`blr-sync-place`	Bidirectional	Syncs locality search selections and pin placements between Map and Studio.
`blr-start-tracker`	Studio → Live Map	Switches view to Live Map and opens the GPS tracking HUD for the selected destination.

6. Summary of Deliverables & System Health

- **Publication PDF:** Compiled directly to `d:\WSZM\SafeRoute_BLR_Deep_Technical_Summary.pdf`.
- **Windows Desktop Executable:** Double-clickable `d:\WSZM\launch_desktop.bat` and `SafeRoute BLR.lnk` shortcut with integrated high-resolution branding.
- **Verification Status:** Python backend compiled with zero errors (`py_compile` exit code 0). Node inline script compilation and DOM ID audits on `bangalore_map.html` and `desktop_ui.html` confirmed zero syntax errors and zero missing DOM element references.